

High-Strength Ultrathin Solid-State Electrolyte with Ion-Rectifying Honeycomb-Like Skeleton for Lithium Metal Batteries Free from External Pressure

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This work breaks the long-standing thickness-performance trade-off in solid-state electrolytes by designing an ultrathin (10 μm) membrane featuring an ion-rectifying honeycomb-like skeleton. The composite skeleton integrates boron nitride nanofibers (BNNFs) and lithiated nafion (Linafion) mosaics, simultaneously enabling rapid Li^+ transport and reinforcing mechanical robustness. The BNNFs provide abundant Lewis acid sites to anchor TFSI $^-$ anions, while Linafion domains facilitate rapid cation exchange via sulfonic acid groups, synergistically promoting Li^+ dissociation from both lithium salts and poly(ester-ether). This unique ion-rectification mechanism achieves ultrafast Li^+ conduction with a low activation energy (0.099 eV), yielding a remarkable room-temperature ionic conductivity of $5.5 \times 10^{-4} \text{ S cm}^{-1}$ and a high Li^+ transference number of 0.85. The all solid-state electrolyte membrane simultaneously exhibits exceptional tensile strength (30 MPa) and compatibility with high-loading NCM811 cathodes (12 mg cm^{-2}) and ultrathin lithium anodes (40 μm). The pouch-type full cells deliver high energy densities (316.6 Wh kg^{-1} , 838.1 Wh L^{-1}) and cycling stability (92.2% retention after 100 cycles) at 25 $^{\circ}\text{C}$. Notably, the full cell can work in a wide temperature range, while maintaining 120.6 mAh g^{-1} at -10 $^{\circ}\text{C}$ without external pressure. This strategy enables practical high-energy solid-state batteries using ultrathin electrolytes with robust mechanics, rapid conduction, and interfacial stability.

1. Introduction

Lithium batteries, as pivotal energy storage devices, have catalyzed revolutions of many modern technologies, including portable electronics, electric vehicles, and unmanned aerial systems, through their continuous performance advancements.^[1] However, achieving simultaneous breakthroughs in energy density and safety remains a critical issue for next-generation lithium batteries.^[2]

Solid-state lithium metal batteries offer a promising solution, as they have the potential to achieve energy density exceeding 500 Wh kg^{-1} while substantially enhancing safety by replacing leak-prone liquid electrolytes.^[3] Among critical components, ultrathin solid-state electrolytes (<20 μm) hold particular promise for minimizing inactive mass and maximizing energy density.^[4–6] However, combining ultrathin architectures with high lithium-ion conductivity, robust mechanical strength, and stable electrode interface remains a tough challenge.^[7]

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DOI: 10.1002/adfm.202511558

Inorganic solid-state electrolytes, while offering high ionic conductivity, suffer from brittleness and a stringent sintering process for the scalable manufacturing of sub-20 μm thin films under cost-effective conditions.^[8] In contrast, polymer electrolytes excel in processability but exhibit mechanical fragility at reduced thicknesses.^[9–11] Hybrid strategies combining mechanically robust substrates with lithium-ion-conductive polymers often sacrifice ionic transport efficiency due to inert substrate materials.^[12,13] For example, Yao et al. achieved a 10 μm polymer electrolyte with 103 MPa tensile strength through the modification of a polyethylene matrix infused with poly(ethylene glycol) methyl ether acrylate and lithium salts. While its limited ionic conductivity ($1.85 \times 10^{-5} \text{ S cm}^{-1}$ at room temperature) necessitated operation at elevated temperatures.^[12] Similarly, Luo et al. fabricated a 4.2 μm bilayer solid-state electrolyte membrane (UFF/PEO/PAN/LiTFSI) with a dual-functional interface for pairing with NCM811 ($\text{LiNi}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}\text{O}_2$), yet it also required operation at 50 $^{\circ}\text{C}$ due to low room-temperature ionic conductivity.^[14] Parallel limitations are also evident in Cui et al.'s 20 μm electrolyte membrane, which was fabricated by infiltrating a polyacrylonitrile (PAN) electrospun matrix with poly(ethylene oxide) (PEO) and lithium salts.^[15] Our prior work demonstrated a 3 μm composite solid-state electrolyte with 29 MPa strength, $1.77 \times 10^{-4} \text{ S cm}^{-1}$ conductivity, and Li^+ transference number of 0.65 at room temperature, by infusing the ester-ether-based lithium-conductive polymers into the mesoporous SiO_2 /PVDF substrate membrane.^[16]

Solid-state electrolytes with ordered ionic transport pathways show enhanced ionic conductivities. Xu et al. incorporated $\text{Yb}_3\text{O}(\text{OH})_6\text{Cl}$ nanowires into a PEO matrix, which generated abundant free Li^+ ions while establishing highly ordered conduction channels, achieving a room-temperature ionic conductivity of $2.67 \times 10^{-4} \text{ S cm}^{-1}$ and a Li^+ transference number of 0.63. However, the membrane thickness was as high as 100 μm .^[17] Jin et al. produced vertically aligned card-house-structured Laponite nanosheets via an ice-templating method and subsequently integrated them into a PEO-LiTFSI matrix to fabricate a 145 μm solid-state electrolyte membrane with an ionic conductivity of $8.9 \times 10^{-4} \text{ S cm}^{-1}$ at 60 $^{\circ}\text{C}$.^[18] However, due to the lack of effective strategies for constructing a continuous and ordered structure with both mechanical and ionic rectification performance, the existing solid-state electrolyte systems still face a trade-off, that is, thickness reduction compromises ionic/mechanical performance, while thicker membranes sacrifice energy density.

Meanwhile, another critical bottleneck lies in unstable interfacial contact between the solid-state electrolyte and the electrodes.^[19,20] At the anode, the strong reducibility of lithium metal triggers parasitic reactions with the electrolyte, escalating interfacial resistance during cycling. Concurrently, the uneven Li^+ flux across the interface promotes lithium dendrite growth and interfacial void formation.^[21] To achieve higher energy density, it is imperative to employ high-loading cathodes and high operation voltage.^[22] However, sluggish Li^+ /electron transport within thick cathodes leads to severe polarization, accelerating electrochemical failure.^[23] Additionally, the anisotropic volume change of nickel-rich layered oxides (e.g., NCM811) will further fracture the cathode-electrolyte interfaces during cycling. The resulting lithium concentration gradients near the interface further accelerate the microcrack formation, resulting in rapid capacity

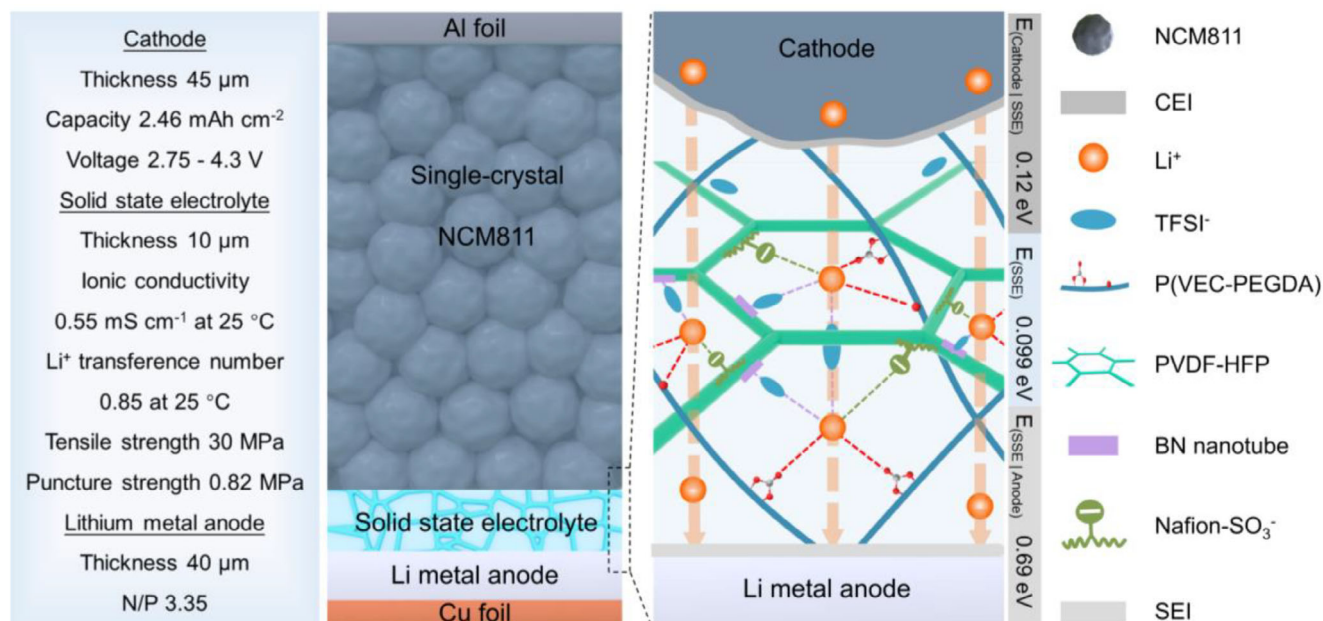
decay.^[24,25] Current strategies, such as introducing lean liquid electrolyte to increase interfacial contact^[26] inevitably induce side reactions between electrodes and the electrolytes, thereby undermining long-term cycling stability.

In this work, an ultrathin solid-state electrolyte membrane was engineered via a multifunctional honeycomb-like skeleton to synergize mechanical reinforcement with ion-rectifying function. As shown in **Figure 1**, the honeycomb-like skeleton is comprised of poly(vinylidene fluoride-co-hexafluoropropylene) (PVDF-HFP) embedded with boron nitride nanofibers (BN-NFs) and lithiated nafion (Linafion) mosaics for Lewis acid and cation-exchange sites, respectively. This architecture demonstrates exceptional mechanical reinforcement and establishes directional Li^+ transport pathways. Subsequently, a Li^+ -conductive poly(ester-ether) copolymer was introduced to form an interpenetrating network structure. The fabricated ultrathin solid-state electrolyte membrane demonstrates a thickness of only 10 μm , with remarkable improvements in room-temperature ionic conductivity ($5.5 \times 10^{-4} \text{ S cm}^{-1}$) and Li^+ transference number (0.85). In comparison, its ionic conductivity and Li^+ transference number have already surpassed those of conventional liquid electrolyte/polyethylene separator systems. The honeycomb architecture effectively distributes external stress, endowing the ultrathin membrane with 30 MPa tensile strength. Coupled with in situ cathode curing, the prepared polymer-based solid-state electrolyte is compatible with thick NCM811 cathodes and thin lithium metal anodes. As a result, the corresponding pouch-type full cell delivers an energy density of 316.6 Wh kg^{-1} (838.1 Wh L^{-1}) and demonstrates exceptional cycling stability under folding, cutting, and low-temperatures conditions. This work explores a strategy transforming passive reinforcement matrices into ion-rectifying active skeletons, thereby break the “thickness-performance” deadlock constraining the ultrathin solid-state electrolytes for practical application.

2. Results and Discussion

2.1. Preparation of Solid-State Electrolytes

Figure 2a illustrates the synthesis process of the functionalized honeycomb-like skeleton and the ultrathin interpenetrating solid-state electrolyte membrane. Initially, the honeycomb-like skeleton PH-BN/LN, composed of poly(vinylidene fluoride-co-hexafluoropropylene) (PVDF-HFP), boron nitride nanofibers (BNNFs), and lithiated Nafion (Linafion), was fabricated using the tape-casting method. As depicted in **Figures S1 and S2** (Supporting Information), the BNNFs have an average length of $\approx 2 \mu\text{m}$ and a diameter of $\approx 200 \text{ nm}$. The optical image of PH-BN/LN shows a uniformly white membrane (**Figure S3a**, Supporting Information), while the scanning electron microscopy (SEM) image highlights its porous honeycomb-like structure (**Figure S3b**, Supporting Information). Moreover, elemental mapping obtained via energy dispersive X-ray spectroscopy (EDS) confirms the homogeneous distribution of Linafion within the PH-BN/LN honeycomb-like skeleton (**Figure S3b**, Supporting Information). The presence of B-N-B peak in the FT-IR spectra (**Figure S4a**, Supporting Information) and the N 1s and B 1s XPS spectra (**Figure S4b,c**, Supporting Information) indicate BN nanofibers



were successfully embedded onto the porous honeycomb-like skeleton.

Subsequently, a precursor solution comprising vinyl ethylene carbonate (VEC), poly(ethylene glycol) diacrylate (PEGDA), and lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) was cast onto the honeycomb-like skeleton and UV-cured to obtain an ultrathin solid-state electrolyte. The wettability of the substrate membrane toward the precursor solution (VEC/PEGDA/LiTFSI) was evaluated through contact angle measurements and uptake tests (Figure 2b; Figure S5, Supporting Information). The precursor solution uptakes of PH, PH-BN, PH-LN, and PH-BN/LN were 285%, 330%, 356% and 378% of the substrate membrane weight, respectively. So, the highest ratio of ionic conductor components and the rapid penetration process should be attributed to the porous honeycomb-like structure of PH-BN/LN (Figure S6, Supporting Information).

The polymerization of the precursor was further analyzed using Fourier transform infrared spectroscopy (FTIR). As shown in Figure 2c, the infrared absorption peak observed $\approx 1650 \text{ cm}^{-1}$ corresponds to the $\text{C}=\text{C}$ bond in VEC.^[16,27] The disappearance of this peak after UV curing confirms the successful completion of the crosslinking polymerization. Additionally, as seen in Figure S7 (Supporting Information), the precursor solution transformed into a non-flowing solid state after UV curing. According to the top-surface and cross-sectional scanning electron microscopy (SEM) images of both the PH-BN/LN substrate membrane and the SSE@PH-BN/LN electrolyte membrane (Figure 2d,e), the SSE@PH-BN/LN membrane achieves a thickness of 10 μm with a smooth surface, indicating effective penetration of the lithium salt-monomers solution into the substrate membrane and subsequent filling the pores after UV curing. Elemental distribution analysis (C, S, F) (Figure S8, Supporting Information) further demonstrates the intimate integration of the PH-BN/LN substrate membrane with the P(VEC-PEGDA) polymer. As shown

in the optical image (Figure 2f), SSE@PH-BN/LN exhibits optical transparency, further confirming the uniformity of the solid-state electrolyte membrane.

Moreover, the thermogravimetric analysis (TGA) and differential thermal analysis (DTA) curves (Figure 2g; Figure S9, Supporting Information) indicate a thermal decomposition onset at 350 $^{\circ}\text{C}$. The thermal stability of PH-BN/LN and SSE@PH-BN/LN is significantly enhanced, which should be attributed to the robust properties of BNNFs and Linafion. Compared to commercial polyethylene (PE) separators (Figure S10, Supporting Information), SSE@PH-BN/LN exhibits superior resistance to thermal shrinkage at 200 $^{\circ}\text{C}$. Furthermore, the SSE@PH-BN/LN membrane shows superior flame retardancy compared to the SSE@PH membrane. Combustion tests reveal that while the SSE@PH membrane ignites after 6 s of flame exposure, the SSE@PH-BN/LN membrane does not combust under the same conditions (Figure S11, Supporting Information). This indicates that the electrolyte has the potential to enhance the safety of lithium-ion batteries by reducing the risk of fire hazards. Meanwhile, the mechanical properties of the electrolyte membranes are also critical to ensure integrity during the manufacturing process and prolonged service life. The puncture resistance of SSE is an important parameter for preventing battery short circuits and lithium dendrite penetration. Figure 2h presents the puncture stress-strain curves for both SSE@PH and SSE@PH-BN/LN, demonstrating that the incorporation of BNNFs and Linafion can significantly enhance the puncture resistance of SSE@PH-BN/LN. To further evaluate mechanical properties, tensile stress-strain tests were performed. Figure 2i, Figures S12, and S13 (Supporting Information) display the stress-strain curves of the substrate membranes and the corresponding solid-state electrolyte membranes. The SSE@PH-BN/LN exhibits exceptional mechanical properties, characterized by a strength of 30 MPa, an elongation of 175% and a calculated Young's modulus of 72.7 MPa. The

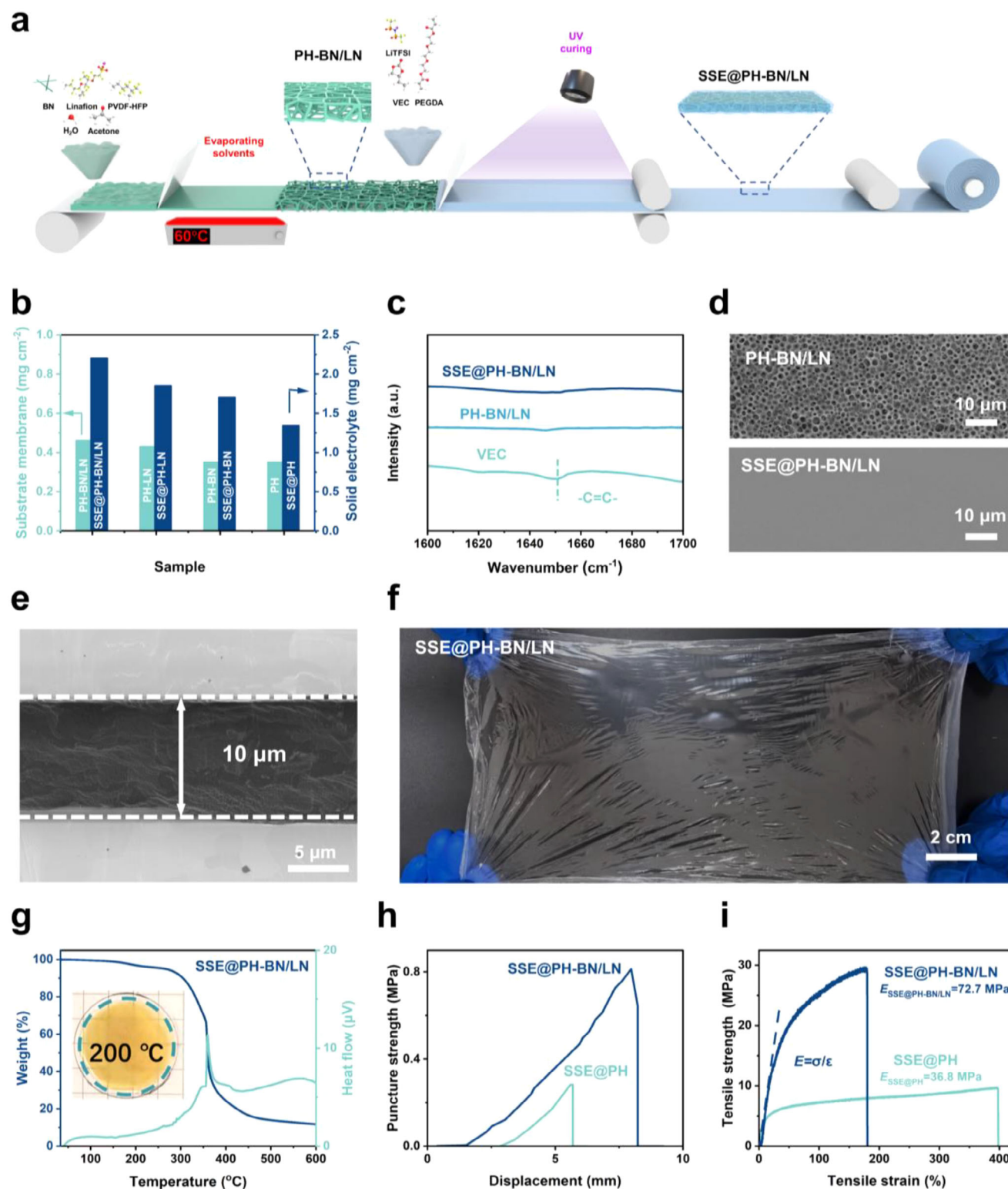


Figure 2. Construction and physical properties of solid-state electrolyte membranes. a) Schematic illustration of the synthetic process for SSE@PH-BN/LN. b) Quality of different substrate membranes and corresponding electrolyte membranes. c) The FTIR spectra of VEC monomer, PH-BN/LN substrate membrane, and SSE@PH-BN/LN electrolyte membrane. d) Top-view SEM images of PH-BN/LN substrate membrane and SSE@PH-BN/LN electrolyte membrane. e) Cross-sectional SEM image of SSE@PH-BN/LN. f) Digital image of SSE@PH-BN/LN. g) DTA and TGA of SSE@PH-BN/LN, the inset shows the thermal shrinkage behavior of SSE@PH-BN/LN at 200 °C for 1 h. h) Puncture strength-displacement curves of SSE@PH-BN/LN and SSE@PH. i) Tensile strength-strain curves of SSE@PH-BN/LN and SSE@PH.

enhanced mechanical performance should be ascribed primarily to the composite honeycomb-like skeleton and the interpenetrating structure. The honeycomb-like architecture can effectively distribute external stress.^[28,29] BNNFs incorporation promotes β -phase crystallization (Figure S4a, Supporting Information), where enriched β -phase domains tighten molecular chain packing to enhance mechanical properties. Furthermore, the combination of BNNFs^[30,31] with flexible Linafon^[32] contributes to the overall mechanical strength of the PH-BN/LN substrate membrane. The introduction of the P(VEC-PEGDA) polymer facilitates the formation of an interpenetrating structure^[33] further reinforcing the mechanical performance of SSE@PH-BN/LN. Collectively, these results illustrate that SSE@PH-BN/LN, constructed from a functionalized honeycomb-like skeleton and an interpenetrating structure, represents a promising solution to address the strength concerns associated with ultrathin solid-state electrolytes.

2.2. Electrochemical Performances of Solid-State Electrolytes

To investigate the influence of the functionalized honeycomb framework on Li^+ transport, a series of electrochemical tests were conducted. Figures 3a and S14 (Supporting Information) display the temperature-dependent ionic conductivity of solid-state electrolyte membranes. At room temperature, the ionic conductivities measured for SSE@PH, SSE@PH-BN, SSE@PH-LN, and SSE@PH-BN/LN were 0.93×10^{-4} , 3.4×10^{-4} , 5.0×10^{-4} , and $5.5 \times 10^{-4} \text{ S cm}^{-1}$, respectively. According to the conductivity relationship (Equations S1 and S2, Supporting Information), the ionic conductivity is positively correlated with the ion concentration. Therefore, higher ion concentrations correspond to enhanced ionic conductivity. The experimental results in Figure 3a agree well with the fact that the honeycomb-like structure aides in the absorption of precursor solutions, thereby increasing the LiTFSI content in the electrolyte membrane.^[34] Furthermore, the incorporation of BN and Linafon facilitates the dissociation of LiTFSI, leading to a higher ion concentration in SSE@PH-BN/LN and corresponding improvement in ionic conductivity, which will be discussed in detail later. The activation energy (E_a) for ion migration can be calculated and fitted using the Arrhenius (Equation S3, Supporting Information) (Figure 3a). Among all samples, SSE@PH-BN/LN exhibited the lowest activation energy of 0.099 eV, indicating a minimal energy barrier for ion transport.

To further elucidate the Li^+ transport capability, the transport behavior of Li^+ and TFSI $^-$ was investigated. The Li^+ transference number (T_{Li^+}), an essential parameter for assessing Li^+ transport efficiency, was determined using the Bruce-Vincent method (Equation S4, Supporting Information) based on the current-time profiles of Li||Li symmetric cells (Figure 3b; Figure S15, Supporting Information). The calculated T_{Li^+} for SSE@PH-BN/LN was 0.85, surpassing those of other samples. This significant enhancement in T_{Li^+} should be attributed to the immobilization of a larger number of anions, coupled with ion-exchange effects, which release more free Li^+ ions. Consequently, SSE@PH-BN/LN exhibits a distinct competitive advantage in comprehensive ionic conductivity and T_{Li^+} compared to other samples, outperforming 1.0 M LiPF_6 , volume ratio EC:DMC:EMC = 1:1:1/9 μm polyethylene separator system (Figure 3b). Specifically, the

room temperature areal conductance (σ_s) of the 10 μm -thick SSE@PH-BN/LN electrolyte was calculated to be 0.56 S cm^{-2} (Equation S5, Supporting Information), in stark contrast to that of the typical liquid electrolytes/separator, which reached only 0.25 S cm^{-2} . In addition, the linear sweep voltammetry (LSV) measurements further revealed the enhanced electrochemical stability of SSE@PH-BN/LN, demonstrating its compatibility with high-voltage cathodes. As illustrated in Figure S16 (Supporting Information), there was no obvious change in current observed at voltage values below 5.5 V, indicating that SSE@PH-BN/LN remains free from redox reactions within high voltage range.

To investigate changes in the coordination environment of TFSI $^-$ and Li^+ , ^7Li solid-state nuclear magnetic resonance (SS NMR) was employed. As shown in Figure 3c, the ^7Li SS NMR spectra of other samples exhibited downfield shifts compared to SSE@PH, with the most pronounced shift observed for SSE@PH-BN/LN. This downfield shift suggests a reduction in electron cloud density around Li^+ , indicative of competitive coordination process involving BN and Linafon.^[35,36] This competitive coordination weakens the interaction between Li^+ and TFSI $^-$ anions. Furthermore, FTIR spectroscopy results agree well with these observations. The characteristic peak $\approx 740 \text{ cm}^{-1}$ in the FTIR spectra,^[27,37] corresponding to the S–N stretching vibration in TFSI $^-$, is sensitive to the formation of ion pairs and can thus be used to analyze the coordination effect of other cations on the S–N bond or other components with TFSI $^-$. As illustrated in Figure S17 (Supporting Information), the proportion of coordinated TFSI $^-$ increased in samples incorporating BN and Linafon compared to SSE@PH, suggesting that a larger number of components engage in coordinating with free-state TFSI $^-$, thereby promoting the migration of free lithium ions.

To further elucidate the coordination environment of TFSI $^-$ and Li^+ , Raman spectroscopy was performed (Figure 3d; Figure S18, Supporting Information). The Raman peak at 737 cm^{-1} corresponds to uncoordinated TFSI $^-$, while the peak at 740 cm^{-1} is attributed to contact ion pairs (CIP) formed by coordinated TFSI $^-$. The peak at 745 cm^{-1} originates from aggregated ion pairs (AGG).^[38,39] The result suggests that while Li^+ coordinates with TFSI $^-$, the Lewis acid sites on BN interact with the S=O bonds of TFSI $^-$, and Linafon competes with TFSI $^-$ for coordination with Li^+ . This competitive coordination weakens the Li^+ coordination environment, facilitating the dissociation of more free Li^+ ions from coordinated TFSI $^-$. This conclusion was further validated by calculating the binding energies between Li^+ and TFSI $^-$, as well as between Li^+ and ethylene carbonate (EC) in P(VEC-PEGDA), in the presence or absence of BN and Linafon.^[36] As shown in Figure 3e and Figure S19 (Supporting Information), the binding energy between Li^+ and TFSI $^-$ was -6.60 eV. However, in the presence of BN and Linafon competing for coordination, the binding energies decreased to -6.29 and -1.85 eV, respectively. Additionally, the binding energy between Li^+ and EC in P(VEC-PEGDA) was -3.74 eV, which further decreased to -1.47 eV when Linafon engaged in competitive coordination.

To understand the competitive coordination effects in SSE@PH-BN/LN, molecular dynamics (MD) simulations were performed for the solid-state electrolytes to characterize the altered coordination environments. Detailed simulation parameters are provided in the supplementary information. The MD

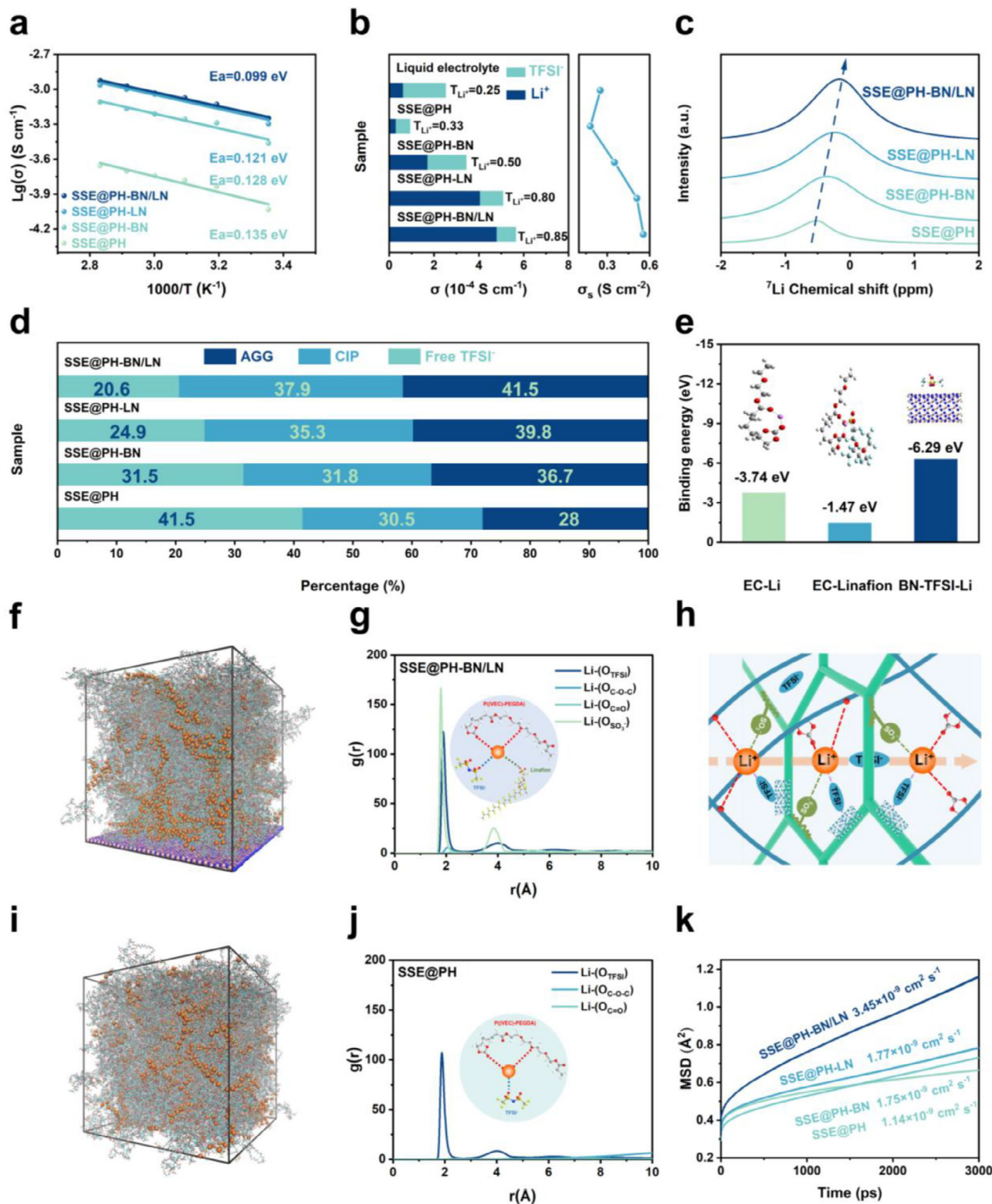


Figure 3. Electrochemical performances and Li⁺ transport mechanism of solid-state electrolyte membranes. a) The Arrhenius plots of different SSEs at various temperatures. b) The ionic conductivities, Li⁺ transference numbers, and areal conductances of different SSEs and liquid electrolyte/separator system (1.0 M LiPF₆, volume ratio EC:DMC:EMC=1:1:1, and 9 μm PE separator) at 25 °C. c) The ⁷Li SSNMR spectra of different SSEs. d) Raman spectra and the proportion of free TFSI⁻, CIP, and AGG in the different SSEs. e) Calculated binding energy of EC-Li, EC-Linafon, and BN-TFSI-Li. Molecular dynamic (MD) snapshots of f) SSE@PH-BN/LN and i) SSE@PH. Calculated radial distribution function (RDF) of g) SSE@PH-BN/LN and j) SSE@PH. h) Schematic illustration of Li⁺ transport in SSE@PH-BN/LN. (k) Calculated mean square displacement (MSD) of Li⁺ in SSEs.

simulation snapshots (Figures 3f,i; Figures S20, S21, Supporting Information) revealed that TFSI⁻ (colored blue) predominantly localizes on the BN surface in SSE@PH-BN/LN, indicating that BN, with its Lewis acid sites, effectively immobilizes TFSI⁻.

Analysis of the radial distribution functions (RDF) and coordination numbers (CNs) (Figure 3g,j; Figures S22, S23, Supporting Information) revealed that compared to SSE@PH, SSE@PH-BN/LN exhibits stronger interactions between Li⁺ and Linafion. Additionally, the CNs for Li-(O_{c-o-c}) and Li-(O_{c=o}) decreased, further indicating that Linafion's role in competitive coordination with Li⁺. Collectively, these findings align with the results from FTIR, Raman, and SS NMR analyses, suggesting that BN and Linafion work synergistically to enhance Li⁺ transport in SSE@PH-BN/LN. Moreover, based on the mean square displacement (MSD) analysis (Figure 3k), the Li⁺ diffusion coefficient (D_{Li^+}) of SSE@PH-BN/LN is determined to be $3.45 \times 10^{-9} \text{ cm}^2 \text{ s}^{-1}$, which is significantly higher than those of other samples (from 1.14×10^{-9} to $1.77 \times 10^{-9} \text{ cm}^2 \text{ s}^{-1}$).

In summary, the high Li⁺ conductivity of SSE@PH-BN/LN with a multifunctionalized honeycomb structure can be attributed to four factors (Figure 3h). First, the honeycomb-like structure enhances the uptake of precursor solution into the substrate membrane. Second, the Lewis acid site of BN adsorbs TFSI⁻, thereby restricting the mobility of TFSI⁻. Third, the cation-exchange effect of Linafion shields the anions and promotes Li⁺ transport. Fourthly, the competitive coordination between BN and Linafion not only enhances the dissociation of Li⁺ but also weakens the coordination of ester and ether with Li⁺, accelerating Li⁺ transport.

2.3. Interfacial Electrochemical Behaviors of Solid-State Electrolyte/Li

In order to evaluate the stability of the interface between the SSE and the lithium metal anode, the lithium stripping/plating behavior was investigated in symmetric cells. During long term cycling (at a current density of 0.5 mA cm^{-2} , and areal capacity of 0.5 mAh cm^{-2}) (Figure 4a), the polarization voltage of Li|SSE@PH|Li cell reached a significant 614 mV, with the cell operating for only 90 h, indicating the instability and incompatibility of the SSE@PH solid-state electrolyte with lithium metal. In contrast, the combined presence of BN and Linafion remarkably decreased the polarization voltage, which stabilized at 170 mV after cycling 1500 h. The lowest resistance value of the Li|SSE@PH-BN/LN|Li cell also confirms the interfacial stability between SSE@PH-BN/LN and lithium metal (Figure S24, Supporting Information).

The impact of different solid-state electrolytes on coulombic efficiency and cycling stability was further evaluated using Li|Cu asymmetric cells, revealing trends consistent with the Li|Li symmetric cell outcomes (Figure S25, Supporting Information). Compared to SSE@PH, the initial coulombic efficiency and cycling stability of the Li|Cu asymmetric cells were obviously improved with the addition of either BN or Linafion. Remarkably, with both BN and Linafion present, the initial coulombic efficiency rose to 69.8%, and after 30 cycles, it reached a stable value $\approx 90.6\%$. These results indicate that BN and Linafion can considerably enhance the reversibility of lithium plating/stripping and

improve the interfacial stability of the electrolyte during cycling with lithium metal anode.

In addition, the SEM image (Figure 4b; Figure S26, Supporting Information) reveals significantly uneven dendritic lithium deposition on lithium anodes cycled with different solid-state electrolytes after 90 h. In contrast, the lithium anode for the Li|SSE@PH-BN/LN|Li cell exhibits a dense, smooth, and dendrite-free deposition layer after long-term cycling. This favorable deposition characteristic can be attributed to the compatibility of SSE@PH-BN/LN with lithium metal anode, alongside the increased Li⁺ transference number (T_{Li^+}).

To further analyze the composition of the solid electrolyte interphase (SEI) on cycled lithium anodes, X-ray photoelectron spectroscopy (XPS) was conducted. The C 1s, F 1s, and N 1s spectra (Figure 4c; Figures S27–S29, Supporting Information) revealed that the SEI predominantly comprises Li₂CO₃, Li₃N, and LiF.^[38,45–47] Notably, the content of Li₃N and LiF is higher in the SEI of the lithium anode contacting SSE@PH-BN/LN. This SEI structure, characterized by excellent Li⁺ conductivity, mechanical robustness, and electronic insulation, facilitates regulating Li⁺ deposition and preventing lithium dendrites growth, thereby prolonging the battery lifespan.

The investigation into the dynamics of lithium deposition also revealed that Li⁺ transport at the SSE/Li interface plays a critical role. The activation energy for Li⁺ transport at the SSE/Li interface was determined via electrochemical impedance spectroscopy (EIS) curves at various temperatures. The results were plotted according to the Arrhenius equation. It is found that SSE@PH-BN/LN exhibited the lowest activation energy of 0.69 eV (Figure 4d). This result indicates that the incorporation of BN and Linafion can modulate the coordination structure of Li⁺, reduce the activation energy for Li⁺ transport, and thus significantly enhance the charge transfer kinetics during lithium deposition.

To confirm the superiority of SSE@PH-BN/LN, the Sand's time-related short-circuit time was tested (Figure S30, Supporting Information), yielding a short-circuit time of $\approx 73 \text{ h}$. Additionally, tests on Li|SSE@PH-BN/LN|Li symmetric cells at varying current densities from 0.1 to 2.0 mA cm^{-2} were performed to determine the critical current density (CCD). The Li|SSE@PH-BN/LN|Li cell demonstrated the highest CCD value. As shown in Figure 4e, although the overpotential increased with rising current density, no short-circuiting was observed until the current density reached 1.5 mA cm^{-2} . Furthermore, the Li|SSE@PH-BN/LN|Li cell maintained a stabler and smaller increase in overpotential during subsequent rate tests (Figure 4f), indicating the excellent electrochemical stability of SSE@PH-BN/LN even under elevated current densities. Moreover, the lithium deposition process was further investigated using the exchange current density derived from lithium symmetric cell tests. The Li|SSE@PH-BN/LN|Li cell exhibits an exchange current density as high as 0.1 mA cm^{-2} (Figure 4g). The elevated exchange current density signifies improved electron transfer at the electrode surface and enhanced Li⁺ diffusion, which is consistent with the high ionic conductivity and Li⁺ transference number of SSE@PH-BN/LN, suggesting accelerated redox reaction rates. These results collectively demonstrate that SSE@PH-BN/LN accelerates reaction rates at the SSE/Li interface while facilitating the lithium plating/stripping process. As depicted in Figure 4h and Table S1

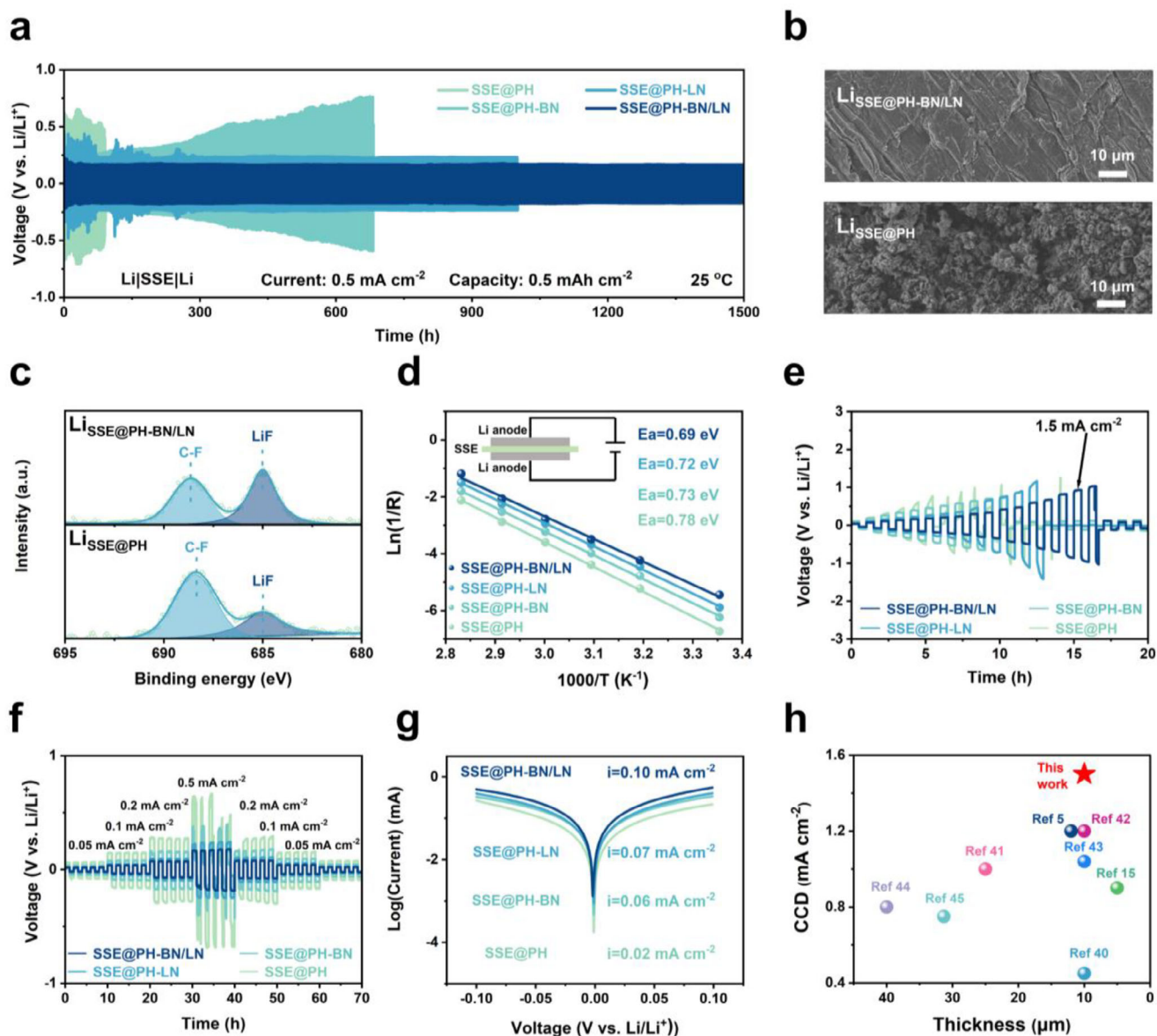


Figure 4. Electrochemical Performance of Symmetric Li||Li Cells a) Long-term cycling performance of Li|SSEs|Li at 0.5 mA cm⁻². b) The surface SEM images of lithium anodes from Li|SSE@PH|Li and Li|SSE@PH-BN/LN|Li after 90 h at 0.5 mA cm⁻². c) F 1s XPS spectra of the cycled Li anodes corresponding to different solid-state electrolytes. d) Interfacial activation energy (E_a) corresponds to the interface impedance in the Li|SSEs|Li symmetric cells. e) Critical current density (CCD) of symmetric Li|SSEs|Li cells. f) Rate tests of symmetric Li|SSEs|Li cells. g) Tafel plots and exchange current density of symmetric Li|SSEs|Li cells. h) Comparisons on CCD and thickness with the contemporary solid-state electrolytes.^[5,12–44]

(Supporting Information), the SSE@PH-BN/LN distinguishes itself among contemporary solid-state electrolytes in terms of CCD and thickness of the solid-state electrolyte membrane.^[5,12–44]

2.4. Interfacial Electrochemical Behaviors of Solid-State Electrolyte/ NCM811

To enhance energy density, it is essential to employ high-loading cathodes. However, as the cathode loading increases, interfacial contact and compatibility issues become more pronounced. In thicker cathodes, the reduced ionic conductivity will lead to exces-

sive polarization, resulting in the electrochemical failure of solid-state batteries. By utilizing SSE@PH-BN/LN electrolyte, characterized by high ionic conductivity and Li⁺ transference number, the cycling of solid-state batteries with high loading cathodes was achieved.

According to measurements from the galvanostatic intermittent titration technique (GITT), the Li⁺ diffusion coefficient in cathodes using SSE@PH-BN/LN electrolyte is higher (Figure 5a). This high ionic conductance and interfacial adaptability facilitates Li⁺ transport within the cathode, yielding high areal capacities. Furthermore, the DC internal resistance of cells with different electrolytes was experimentally measured.^[48]

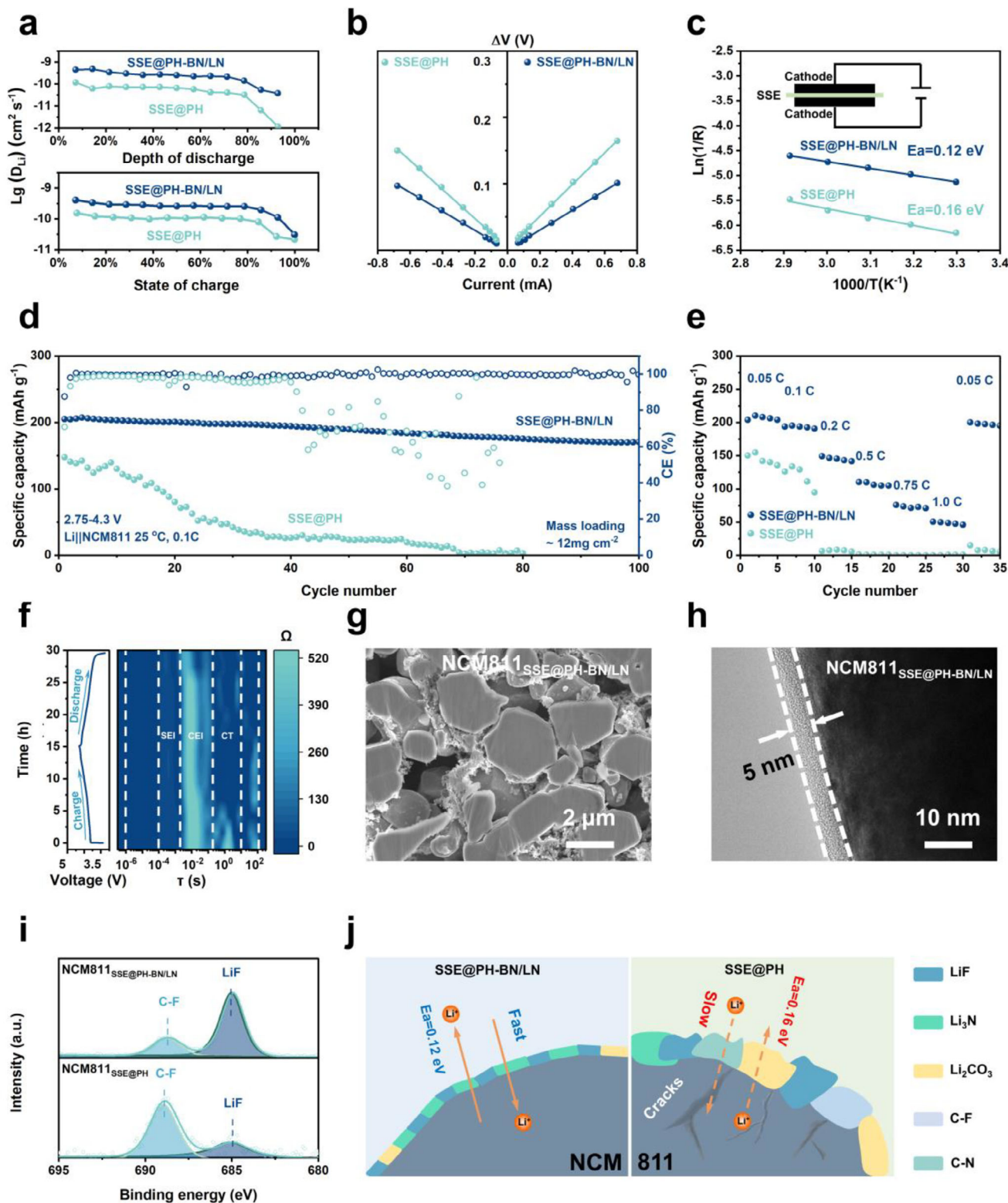


Figure 5. Electrochemical Performance of Li||NCM811 Cells a) Li^+ ions diffusion coefficients calculated from the GITT curves. b) The DC-IR experiments of Li|SSEs|NCM811. c) Interfacial activation energy (E_a) corresponds to the interface impedance in the NCM811|SSEs|NCM811 symmetric cells. d) Cycling stability of Li|SSEs|NCM811 coin-type full cells at 0.1 and 25 °C. e) Rate performance of Li|SSEs|NCM811 from 0.05 to 1 at 25 °C. f) DRT profile transformation from impedance evolution of Li|SSE@PH-BN/LN|NCM811. g) SEM image of NCM811 contacting SSE@PH-BN/LN after 80 cycles. h) TEM images of NCM811 contacting SSE@PH-BN/LN after 80 cycles. i) F 1s XPS spectra of NCM811 contacting SSE@PH-BN/LN and SSE@PH respectively after 80 cycles. j) Schematic illustration of CEI layer on cycled NCM811 particles in Li|SSE@PH|NCM811 and Li|SSE@PH-BN/LN|NCM811.

The voltage responses of the cells are illustrated in Figure S31 (Supporting Information). The internal resistance during charge/discharge was determined via linear regression analysis of the voltage-current variation (ΔV -i) plots (Figure 5b). Cells employing SSE@PH-BN/LN exhibits a relatively lower slope in the ΔV -i plot, indicating reduced internal resistance in the Li|SSE@PH-BN/LN|NCM811 cell due to the superior ionic conductivity and Li^+ transference number of SSE@PH-BN/LN.

By constructing NCM811|NCM811 symmetric cells and analyzing their resistances at various temperatures, the activation energy was calculated based on the thermal activation equation. Figure 5c illustrates the relationship between $\log(1/R)$ and $1000/T$, where SSE@PH-BN/LN shows a smaller absolute slope, corresponding to a calculated interfacial activation energy of 0.12 eV. The lower activation energy suggests a diminished energy barrier for Li^+ transport across the cathode electrolyte interphase (CEI).

The solid-state electrolytes were further evaluated in coin-type full cells, coupling with the high-loading NCM811 cathodes (12 mg cm^{-2}) and the thin lithium anodes ($40 \mu\text{m}$). At room temperature, the cell using the SSE@PH electrolyte exhibited a sharp capacity decay, with capacity nearing zero after 80 cycles. This severe capacity degradation should be attributed to sluggish Li^+ transport in the high-loading cathode. In contrast, cells incorporating either BNNFs or Linafion showed improved capacity and capacity retention (Figure S32, Supporting Information). The simultaneous incorporation of both BN and Linafion into the electrolyte further enhanced the cell capacity and cyclability, maintaining a capacity of 170.5 mAh g^{-1} after 100 cycles, with a retention of 83.2% (Figure 5d). As demonstrated in Figure 5e, the discharge capacities at current densities of 0.05, 0.1, 0.2, 0.5, 0.75, and 1 C were 204.1, 195.1, 145.5, 105.1, 73.7, and 48.8 mAh g^{-1} , respectively. Notably, when the current density was restored to 0.05 C, the specific discharge capacity recovered, confirming the favorable rate capability of the Li|SSE@PH-BN/LN|NCM811 cell. Furthermore, the Li|SSE@PH-BN/LN|NCM811 cell maintains exceptional temperature-adaptive cyclability, operating effectively at -10°C while achieving 88.1% capacity retention after 100 cycles at 60°C (Figure S33, Supporting Information). Notably, when paired with high-loading LiFePO_4 cathodes, it delivers 95.1% capacity retention over 100 stable cycles (Figure S34, Supporting Information).

To investigate the interfacial changes between the solid-state electrolyte and the cathode, in situ impedance measurements were performed, deriving the distribution of relaxation times (DRT) from a series of impedance variations. Figure 5f and Figure S35 (Supporting Information) display the voltage profiles during charging and discharging, along with the corresponding DRT. A charge/discharge rate of 0.1 C was applied, and impedance measurements were conducted at various voltage points during cycling. The frequency range for impedance testing was $0.01\text{--}10^6 \text{ Hz}$, reflecting the stability of Li^+ transport throughout the charge/discharge process. The peaks at $10^{-6}\text{--}10^{-4} \text{ s}$ correspond to the resistance of the solid-state electrolyte (R_s), while those at $10^{-4}\text{--}10^{-3} \text{ s}$, $10^{-3}\text{--}10^{-1} \text{ s}$ and $10^{-1}\text{--}10 \text{ s}$ correspond to the resistance at the lithium anode/SSE interphase (R_{SEI}), the NCM811/SSE interphase (R_{CEI}) and the charge transfer resistance at the interface (R_{CT}), respectively. The peak at $10\text{--}10^2 \text{ s}$ corresponds

to solid-state diffusion within the cathode.^[49–54] Notably, the Li|SSE@PH-BN/LN|NCM811 cell exhibited lower peak intensity for R_{SEI} and R_{CEI} than Li|SSE@PH|NCM811, indicating faster Li^+ transport in the SSE@PH-BN/LN/electrodes interphases. Additionally, the R_{CT} peak for the Li|SSE@PH-BN/LN|NCM811 cell was also lower than that of Li|SSE@PH|NCM811, suggesting that the stable contact between NCM811 cathode and SSE@PH-BN/LN facilitates the charge transfer process. Furthermore, we compared the impedance of Li|SSE@PH-BN/LN|NCM811 and Li|SSE@PH|NCM811 cells at the 1st and 80th cycles. Electrochemical impedance spectroscopy (EIS) results (Figure S36, Supporting Information) reveal significantly lower interfacial resistance in Li|SSE@PH-BN/LN|NCM811 versus Li|SSE@PH|NCM811. Moreover, after 100 cycles, the Li|SSE@PH-BN/LN|NCM811 cell exhibits minimal impedance growth. The results are in consistent with the lower interfacial activation energies for SSE@PH-BN/LN.

To understand the surface and structural stability of the NCM811 cathode during cycling, SEM, high-resolution transmission electron microscopy (HRTEM), and XPS analyses on the NCM811 cathode were conducted. Figure S37 (Supporting Information) shows structural damage and microcracks in the NCM811 cathode after 80 cycles in the Li|SSE@PH|NCM811 cell. The thick ($\approx 10 \text{ nm}$) and non-uniform CEI will aggravate the Li^+ concentration gradient near the interface and accelerate the formation of microcracks (Figure S38, Supporting Information). Conversely, no microcracks were observed in the NCM811 cathode in the Li|SSE@PH-BN/LN|NCM811 cell (Figure 5g). These beneficial effects should be attributed to the formation of a uniform and thin CEI layer ($\approx 5 \text{ nm}$) (Figure 5h). Furthermore, the XPS analysis of the CEI (Figure 5i; Figures S39, S40, Supporting Information) from the Li|SSE@PH|NCM811 cell indicated a substantial ratio of Li_2CO_3 . In contrast, the CEI from the Li|SSE@PH-BN/LN|NCM811 cell exhibits less Li_2CO_3 but more LiF and Li_3N . These results demonstrate that the formation of an inorganic CEI layer enriched with LiF and Li_3N effectively facilitates Li^+ transport and protects the cathode.^[38,55,56]

Based on the above discussions, Figure 5j illustrates the effectiveness of SSE@PH-BN/LN as a solid-state electrolyte in Li|NCM811 batteries. The high ionic conductivity and high Li^+ transference number associated with SSE@PH-BN/LN not only facilitate Li^+ transport but also promote the formation of a thin and uniform CEI layer, enriched with LiF and Li_3N . This CEI effectively suppresses interfacial side reactions between the cathode and the solid-state electrolyte, as well as cracks in the cathode.

2.5. Performances of Li|NCM811 Pouch Cells

To further demonstrate the feasibility of solid-state electrolytes for practical applications, the Li|SSE@PH-BN/LN|NCM811 pouch-type full cell was designed for various practical scenarios, safety evaluations, and cycling performance tests under multiple temperature conditions. The pouch cell consists of a $40 \mu\text{m}$ lithium foil, a $10 \mu\text{m}$ SSE@PH-BN/LN solid-state electrolyte membrane and a high-loading single-crystal NCM811 cathode (12 mg cm^{-2}) (Figure 6a). The cell demonstrates an impressive initial discharge specific capacity of 206.5 mAh g^{-1} at

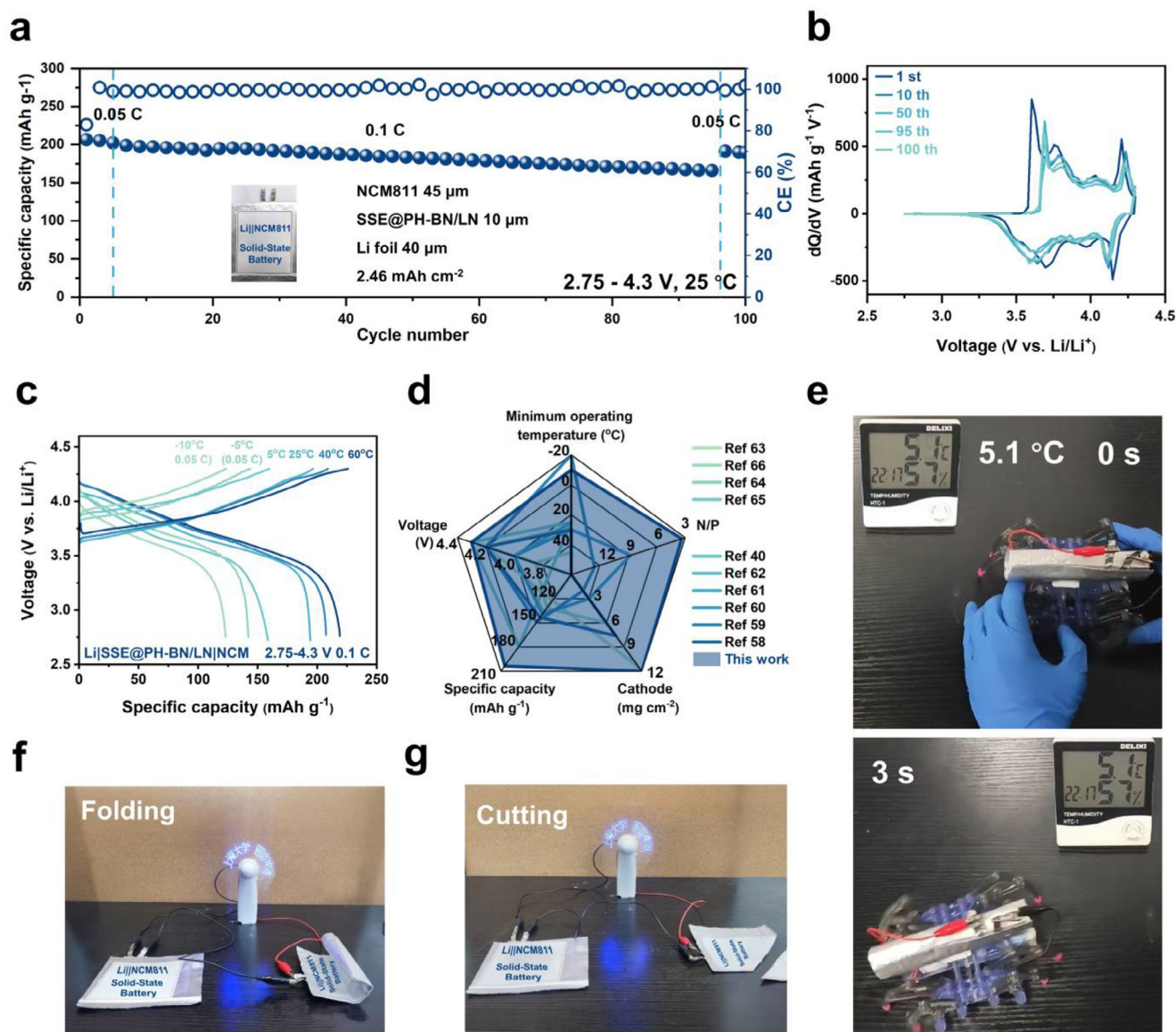


Figure 6. Performances of the Li||NCM811 pouch cells a) Cycling performance of the Li||SSE@PH-BN/LN|NCM811 pouch cell. b) The dQ/dV curves of the Li||SSE@PH-BN/LN|NCM811 pouch cell at different cycles. c) Charge/discharge curves of the Li||SSE@PH-BN/LN|NCM811 pouch cell at -10 to 60 °C. d) Comparison of the Li||SSE@PH-BN/LN|NCM811 cell with other SSE-based cells in terms of minimum operating temperature, N/P ratio, cathode loading, specific capacity, and operating voltage.^[12,59–67] e) Digital photographs of a crawling spider robot powered by the Li||SSE@PH-BN/LN|NCM811 pouch cell at 5.1 °C. f, g) Li||SSE@PH-BN/LN|NCM811 pouch cells powering a LED fan under folding and cutting conditions.

0.05 C. After undergoing 90 cycles at 0.1 C, the specific capacity retains a value of 165.8 mAh g⁻¹. Notably, when the current density returns to 0.05 C after 95 cycles, the specific capacity recovers to 189.6 mAh g⁻¹. After 100 cycles, the capacity retention is 92.2%. The differential capacity (dQ/dV) curves during cycling (Figure 6b) confirm that the Li||SSE@PH-BN/LN|NCM811 pouch cell exhibits excellent reversibility and structural stability.

Moreover, the Li||SSE@PH-BN/LN|NCM811 pouch cell exhibits impressive rate performance, achieving specific capacities of 205.3, 198.4, 137.6, and 99.7 mAh g⁻¹ at stepwise current densities of 0.05, 0.1, 0.2, and 0.5 C (Figure S41, Supporting Information). To evaluate the extensive operational temperature range, the performance test on the Li||SSE@PH-BN/LN|NCM811 cell

was conducted across temperatures from -10 to 60 °C (Figure 6c). The cell exhibited a capacity of 218.9 mAh g⁻¹ at 60 °C, and it still delivered a commendable capacity of 120.6 mAh g⁻¹ at -10 and 0.05 C without external pressure. In contrast, the sulfide-based solid electrolytes usually require an external pressure ranging from 60 to 600 MPa to achieve low-temperature performance.^[57,58]

Figure 6d and Table S2 (Supporting Information) provide a comparative analysis of the Li||SSE@PH-BN/LN|NCM811 cell with the previous reports, highlighting its favorable comprehensive performance in terms of minimum operating temperature, low N/P ratio, high cathode loading, and high operating voltage.^[12,59–67] The outstanding battery performance should be

attributed to the effective construction of Li^+ transport pathways within the solid-state electrolyte, as well as the formation of stable CEI and SEI.

Remarkably, the pouch cell demonstrates the ability to power a spider robot to crawl rapidly at 5.1 °C (Figure 6e; Video S1, Supporting Information), showcasing its exceptional performance across a wide temperature range. Furthermore, the robustness of the pouch cell under adverse conditions has also been verified. As shown in Figure 6f,g, the pouch cell was able to power an LED fan even after being subjected to folding and cutting, highlighting the potential of SSE@PH-BN/LN solid-state electrolytes in flexible pouch cell applications.

3. Conclusion

In summary, this work demonstrates a strategy for ultrathin solid-state electrolytes (SSEs) through a rationally designed multifunctional honeycomb-like skeleton. The integration of boron nitride nanofibers (BNNFs) and lithiated Nafion (Linafion) mosaics creates an ion-rectifying nanoarchitecture that synergistically tackles the critical trade-offs between thickness, ion transport, and mechanical robustness.

The Lewis acid sites of BNNFs immobilize TFSI⁻ anions, while the sulfonic groups of Linafion facilitate cation exchange, cooperatively liberating Li^+ ions from both LiTFSI and poly(ester-ether). This unique mechanism achieves a ultralow activation energy (0.099 eV), enabling ultrahigh ionic conductivity ($5.5 \times 10^{-4} \text{ S cm}^{-1}$) and an exceptional Li^+ transference number (0.85). Meanwhile, the honeycomb skeleton endows the ultrathin membrane (10 μm) with robust tensile strength (30 MPa) and puncture resistance, overcoming the fragility typical of sub-20 μm SSEs. When paired with high-loading NCM811 cathodes (12 mg cm^{-2}) and ultrathin Li anodes (40 μm), pouch cells deliver record energy densities (316.6 Wh kg^{-1} , 838.1 Wh L^{-1}) and 92.2% capacity retention after 100 cycles. Crucially, the cells operate without external pressure even at -10 °C (120.6 mAh g^{-1}), showing superior low-temperature performance.

This work transforms passive reinforcement matrices into ion-regulating active frameworks, providing a scalable method for high-energy, pressure-free all-solid-state batteries. In summary, an ultrathin solid-state electrolyte membrane with remarkable electrochemical properties and mechanical performance was constructed by incorporating a multifunctional honeycomb-like skeleton.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

The authors acknowledge the Natural Science Foundation of Shanghai (23ZR1423600), National Key R&D Plan of the Ministry of Science and Technology of China (2022YFE0122400), Science and Technology Commission of Shanghai Municipality (20dz1201102), Professional and Technical Service Platform for Designing and Manufacturing of Advanced Composite Materials (Shanghai), Engineering Research Center of Material Composition and Advanced Dispersion Technology, Ministry of Education.

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

honeycomb-like skeleton, ion-rectification, lithium metal batteries, solid-state electrolyte, ultrathin

Received: June 10, 2025

Revised: June 21, 2025

Published online:

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